



Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: B.Tech., III & CSBS

Course Code & Title: AD3351 Design and Analysis of Algorithms

Name of the Faculty member: Ms.K.Usharani AP/CSBS

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Innovative Practice Description

- **Unit / Topic:** Unit 1 / Visualization of Algorithms
- **Course Outcome:** CO 1 – Analyze the efficiency of recursive and non-recursive algorithms mathematically
- **Topic Learning Outcome:** Analyzing recursive and non- recursive algorithms in different methods.
- **Activity Chosen:** Peer Teaching
- **Justification:**
 - Peer Teaching fosters teamwork, cooperation, patience, and better social skills. In a cooperative peer learning environment, each student's strengths can serve to complement the group and enhance learning. Teaching by their peers will improve the student's understanding.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 1. **Materials used :**
 - a. Textbook - Anany Levitin, "Introduction to the Design and Analysis of Algorithms".
 - b. Visualization sites:
 - i. <https://algorithm-visualizer.org/>
 - ii. <https://visualgo.net/en>
 - iii. <https://www.cs.usfca.edu/~galles/visualization/ComparisonSort.html>
 - iv. <https://www.sortvisualizer.com/>
 2. **Prerequisite :** Basic knowledge on sorting algorithms.
 3. All students were asked to discuss in their teams on the given topic with the help of textbook.
 4. 10 Interested students are asked to explain the concepts to their class members.
 5. To check their knowledge on this topic, questions were asked to students.
 6. After that, the faculty consolidated the concepts once again to students and the answers for the questions were also discussed.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO9	PO10	PO12	PSO2
CO5	2	1	3	3	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PO12	PSO2
	2	1	3	3	2	2
Justification for correlation	Apply the knowledge of engineering fundamentals like algorithms	Identify and analyze complex engineering problems related to algorithms	Function effectively as an individual and as a member in a team	Communicate effectively on complex engineering activities with engineering community.	Recognize the need and ability to engage in independent and life-long learning.	Exhibit proficiency in software development.

- **Images / Screenshot of the practice:**





- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students felt that they can able to understand and get clarity on the topic very easily by sharing their knowledge with their friends. It improves their self-learning ability.

- ❖ ***Benefit of the practice:***

Students were actively participated in this activity.

Students interaction with their peer members and the class during the activity was good and effective.

- ❖ ***Challenges faced in implementation:***

Only few students are ready to present the topic in front of their class.

Some students are hesitated to share their knowledge in the class. And they are motivated to come out of their fear.

References:

1. <https://www.wgu.edu/blog/peer-learning2208.html#close>